BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: 92101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. _____ is a vital process in the life of plants.

(A) Reproduction
(B) Translocation
(C) Transpiration
(D) Locomotion

Q2. White blood cells are called.

(A) Erythrocytes
(B) Leucocytes
(C) Thrombocytes
(D) Phagocytes

Q3. Plants require _____ to form chlorophyll.

(A) Nitrogen
(B) Magnesium
(C) Calcium
(D) Iron

Q4. Krebs cycle occur in.

(A) Mitochondria
(B) Cytoplasm
(C) Nucleus
(D) Ribosomes

Q5. _____ is the study of transformation of energy in organisms.

(A) Photosynthesis
(B) Respiration
(C) Transpiration
(D) Bioenergetics

Q6. The building up of larger molecules from smaller ones is known as.

(A) Anabolism
(B) Catabolism
(C) Metabolism
(D) Canabolism

Q7. Pairing up of homologous chromosomes is called.

(A) Synapsis
(B) Crossing over
(C) Chiasmata
(D) Disjunction

Q8. Replication of DNA starts in.

(A) G0 phase
(B) G1 phase
(C) G2 phase
(D) S phase

Q9. _____ tissue are made up of those cells which have the ability to divide.

(A) Meristematic
(B) Conducting
(C) Ground
(D) Supporting

Q10. The movement of substances against the concentration gradient is known as.

(A) Osmosis
(B) Diffusion
(C) Plasmolysis
(D) Active Transport

Q11. _____ is the method by which biologists divide organisms into groups and subgroups on the basis of similarities and differences.

(A) Catagarization
(B) Classification
(C) Organization
(D) Distribution

Q12. The study of transmission of characters from parents to offspring is called.

(A) Genetics
(B) Ecology
(C) Taxonomy
(D) Morphology

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Write short note on the levels of Biological organization.

(ii) How is plasmodium transmitted to human beings?

(iii) What is mitochondria? Briefly describe its structure and draw its diagram.

(iv) Write short note on facilitated diffusion.

(v) What is cytokinesis? How this process occur in animal cell?

(vi) What are cofactors? Briefly describe their types.

(vii) Briefly describe dark reaction as a step of mechanism of Photosynthesis.

(viii) Briefly describe ATP as the energy currency of cell.

(ix) What is malnutrition? Why it is considered to be a health hazard?

(x) What is transpiration? How the rate of transpiration is affected by light and temperature?

(xi) What is pulse? Briefly describe why pulse rate is an easy way to understand the condition of heart?

SECTION "C"

Maximum Marks: 21

> Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [3] Explain hierarchy of taxonomy.

(b) [4] What are phloem tissues? Differentiate between sieve tube and companion cell. Also draw their diagrams

QIV.

(a) [3] Briefly describe the advantages of binomial nomenclature.

(b) [4] Briefly describe lactic acid fermentation.

QV.

(a) [3] Briefly describe intracellular and extracellular enzymes.

(b) [4] Briefly describe the significance of mitosis.

QVI.

(a) [3] Write short note on Proteins.

(b) [4] Describe and draw the internal structure of human heart.